📌 Objective

To perform basic network scanning using **Nmap** and understand different scan types such as ping scan, port scan, and service version detection in a controlled lab environment.

Basic Concepts (Theory)

🔹 What is Nmap?

Nmap (Network Mapper) is a tool used to discover devices on a network and identify open ports and running services. It is widely used in cybersecurity and network administration.

🔹 What is Network Scanning?

Network scanning is the process of identifying active devices, open ports, and services in a network.

🔹 What is an IP Address?

An IP address is a unique identifier assigned to each device on a network.
Example: 192.168.1.10

🔹 What is a Port?

A port is a communication endpoint used by services.
Example:

- Port 80 → HTTP (web)
- Port 22 → SSH (remote login)

🔹 What is a Service?

A service is a program running on a port.
Example: Apache web server running on port 80.

🔹 What is a Flag?

A flag is an option used with a command to change its behavior.
Example: -sn, -sV

🔹 What is a Target?

The system or device that is being scanned.
Example: 192.168.1.5

🔹 What is a Subnet?

A subnet is a range of IP addresses in a network.
Example: 192.168.1.0/24

🔧 **Practical Implementation**

🔹 **Environment Setup**

- **Operating System:** Kali Linux
- **My IP Address:** 192.168.1.10
- **Target IP:** 192.168.1.5
- **Network Range:** 192.168.1.0/24

Step 1: Ping Scan (Find Live Devices)

Command I Used

nmap -sn 192.168.1.0/24

What I Learned

- nmap → tool for scanning networks
- -sn → flag to only check which hosts are alive (no port scan)

Output I Got (Example)

```
Nmap scan report for 192.168.1.1
Host is up

Nmap scan report for 192.168.1.5
Host is up

Nmap scan report for 192.168.1.10
Host is up
```

My Observation

I found 3 active devices:

- Router → 192.168.1.1
- Target system → 192.168.1.5
- My Kali machine → 192.168.1.10

Step 2: Port Scan

Command I Used

nmap 192.168.1.5

What I Learned

- This scans the most common 1000 ports
- Helps to find open services

Output I Got

```
PORT    STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
443/tcp  open  https
```

My Observation

- Port 22 → SSH (remote login)
- Port 80 → Website (HTTP)
- Port 443 → Secure website (HTTPS)

Step 3: Service Version Scan

Command I Used

nmap -sV 192.168.1.5

What I Learned

- -sV → shows version of services
- Useful for finding vulnerabilities

Output I Got

```
PORT    STATE SERVICE VERSION
22/tcp   open  ssh    OpenSSH 8.2
80/tcp   open  http   Apache 2.4.41
443/tcp  open  https  Apache 2.4.41
```

My Observation

- Found exact versions of services
- This information can help in security testing

Summary Table

Step	Command	Purpose
1	nmap -sn 192.168.1.0/24	Find live devices
2	nmap 192.168.1.5	Find open ports
3	nmap -sV 192.168.1.5	Find service versions

Conclusion

In this lab, I learned how to use Nmap for basic network scanning. First, I found live systems using a ping scan. Then I scanned open ports on a target machine. Finally, I checked the versions of services running on those ports. This helped me understand how attackers and security professionals gather information about a system.